

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: 5

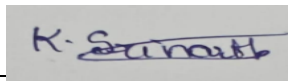
Total Marks: 50

Annexure A

Code of Conduct:

I K.Srinath(192524220) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

A rectangular box containing a handwritten signature in blue ink that reads "K. Srinath".

Question 1 — RSA Key Generation and Security Analysis

Problem Understanding

This problem requires generation of an RSA public–private key pair from two given prime numbers, followed by encryption and decryption of a plaintext message to verify the correctness of the RSA algorithm.

- Given: $p = 17$, $q = 19$
- Public exponent chosen: $e = 5$
- Message to be encrypted: $M = 7$
- Required: n , $\phi(n)$, private key d , ciphertext C , and recovered message M

Method / Formulae Used

$$n = p \times q \quad (1)$$

$$\phi(n) = (p - 1) \times (q - 1) \quad (2)$$

$$e \times d \equiv 1 \pmod{\phi(n)} \quad (3)$$

$$C = M^e \bmod n \quad (\text{Encryption}) \quad (4)$$

$$M = C^d \bmod n \quad (\text{Decryption}) \quad (5)$$

Solution

Step 1 — Compute the modulus n

$$n = p \times q = 17 \times 19 = 323$$

Step 2 — Compute Euler's totient $\phi(n)$

$$\phi(n) = (p - 1)(q - 1) = (17 - 1)(19 - 1) = 16 \times 18 = 288$$

Step 3 — Compute the private key d

d must satisfy: $e \times d \equiv 1 \pmod{288}$, with $e = 5$

$$5 \times 173 = 865 = (288 \times 3) + 1 \Rightarrow 865 \bmod 288 = 1$$

Therefore $d = 173$

Step 4 — Encrypt the message $M = 7$

$$C = M^e \bmod n = 7^5 \bmod 323$$

Sub-computation	Working	Result (mod 323)
7^2	$7 \times 7 = 49$	49
7^4	$49 \times 49 = 2401 = 323 \times 7 + 140$	140
7^5	$140 \times 7 = 980 = 323 \times 3 + 11$	11

Ciphertext: $C = 11$

Step 5 — Decrypt the ciphertext $C = 11$

$$M = C^d \bmod n = 11^{173} \bmod 323$$

Using fast (square-and-multiply) exponentiation, repeated squaring of 11 modulo 323 and combining the terms corresponding to the binary form of 173 ($173 = 10101101_2$) yields:

Power of 11 (mod 323)	Value
11^1	11
11^2	121
11^4	246
11^8	290
11^{16}	155
11^{32}	192
11^{64}	184
11^{128}	289

$173 = 128 + 32 + 8 + 4 + 1$, so:

$$M = (11^{128} \times 11^{32} \times 11^8 \times 11^4 \times 11^1) \bmod 323 = (289 \times 192 \times 290 \times 246 \times 11) \bmod 323 = 7$$

The decrypted message $M = 7$ matches the original plaintext, confirming correct RSA operation.

Final Answer

$n = 323$, $\phi(n) = 288$, $d = 173$, Ciphertext $C = 11$, Decrypted $M = 7$ (verified)

Analysis

Choosing very small primes such as $p = 17$ and $q = 19$ makes RSA cryptographically weak because $n = 323$ can be factorised almost instantly by trial division, exposing both $\phi(n)$ and the private key d . In practical systems, p and q must each be at least 1024 bits so that factoring n is computationally infeasible. This example is useful only for illustrating the algorithm; it offers no real-world security. Secure RSA deployments therefore rely on large, randomly generated primes and key sizes of 2048 bits or more.

Question 2 — ElGamal Encryption Analysis

Problem Understanding

This problem requires computing the ElGamal public key, encrypting a message using a random value k , and then decrypting the resulting ciphertext pair to recover the original message.

- Given: $p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
- Required: public key y , ciphertext pair $(C1, C2)$, and recovered message M

Method / Formulae Used

$$y = g^x \bmod p \quad (\text{Public key}) \quad (1)$$

$$C1 = g^k \bmod p \quad (2)$$

$$C2 = (M \times y^k) \bmod p \quad (3)$$

$$M = (C2 \times (C1^x)^{-1}) \bmod p \quad (\text{Decryption}) \quad (4)$$

Solution

Step 1 — Compute the public key y

$$y = g^x \bmod p = 5^7 \bmod 23$$

Sub-computation	Working	Result (mod 23)
5^2	$25 = 23 + 2$	2
5^4	$2^2 = 4$	4
5^6	$5^4 \times 5^2 = 4 \times 2 = 8$	8
5^7	$8 \times 5 = 40 = 23 + 17$	17

Public key: $y = 17$

Step 2 — Compute $C1$

$$C1 = g^k \bmod p = 5^3 \bmod 23 = 125 \bmod 23 = 10$$

Step 3 — Compute $C2$

First compute $y^k \bmod p = 17^3 \bmod 23$

$$17^2 = 289 \bmod 23 = 13; \quad 17^3 = 13 \times 17 = 221 \bmod 23 = 14$$

$$C2 = (M \times y^k) \bmod p = (10 \times 14) \bmod 23 = 140 \bmod 23 = 2$$

Ciphertext pair: $(C1, C2) = (10, 2)$

Step 4 — Decrypt the ciphertext

$$\text{Compute } s = C1^x \bmod p = 10^7 \bmod 23$$

Sub-computation	Working	Result (mod 23)
10^2	$100 = 92 + 8$	8

10^4	$8^2 = 64 = 46 + 18$	18
10^6	$18 \times 8 = 144 = 138 + 6$	6
10^7	$6 \times 10 = 60 = 46 + 14$	14

$s = 14$. Its modular inverse mod 23: since $14 \times 5 = 70 = 69 + 1$, $s^{-1} = 5$

$M = (C_2 \times s^{-1}) \bmod p = (2 \times 5) \bmod 23 = 10$

The recovered message $M = 10$ matches the original plaintext.

Final Answer

Public key $y = 17$, Ciphertext $(C_1, C_2) = (10, 2)$, Decrypted $M = 10$ (verified)

Analysis

The random value k is the key to ElGamal's security: it makes the encryption probabilistic rather than deterministic, so encrypting the same message twice produces different ciphertexts. This prevents an attacker from recognising repeated messages or performing simple pattern analysis. However, k must be freshly generated for every encryption and never reused, since reuse across two messages allows an attacker to eliminate the random term and recover the private key. ElGamal's security rests on the computational difficulty of the discrete logarithm problem in the chosen group.

Question 3 — RSA Digital Signature Verification

Problem Understanding

This problem requires generating an RSA key pair, signing a message using the private key, and verifying the signature using the corresponding public key.

- Given: $p = 11$, $q = 17$, $e = 7$, message $M = 8$
- Required: n , $\phi(n)$, private key d , digital signature S , and verification result

Method / Formulae Used

$$n = p \times q \quad (1)$$

$$\phi(n) = (p - 1)(q - 1) \quad (2)$$

$$e \times d \equiv 1 \pmod{\phi(n)} \quad (3)$$

$$S = M^d \bmod n \quad (\text{Signature generation}) \quad (4)$$

$$M' = S^e \bmod n \quad (\text{Signature verification}) \quad (5)$$

Solution

Step 1 — Compute the modulus n

$$n = p \times q = 11 \times 17 = 187$$

Step 2 — Compute Euler's totient $\phi(n)$

$$\phi(n) = (11 - 1)(17 - 1) = 10 \times 16 = 160$$

Step 3 — Compute the private key d

d must satisfy: $e \times d \equiv 1 \pmod{160}$, with $e = 7$

$$7 \times 23 = 161 = (160 \times 1) + 1 \Rightarrow 161 \bmod 160 = 1$$

Therefore $d = 23$

Step 4 — Generate the digital signature

$$S = M^d \bmod n = 8^{23} \bmod 187$$

Sub-computation	Working	Result (mod 187)
8^2	64	64
8^4	$64^2 = 4096 = 187 \times 21 + 169$	169
8^8	$169^2 = 28561 = 187 \times 152 + 137$	137
8^{16}	$137^2 = 18769 = 187 \times 100 + 69$	69

$23 = 16 + 4 + 2 + 1$, so:

$$S = (8^{16} \times 8^4 \times 8^2 \times 8^1) \bmod 187 = (69 \times 169 \times 64 \times 8) \bmod 187 = 83$$

Digital signature: $S = 83$

Step 5 — Verify the signature

$$M' = S^e \bmod n = 83^7 \bmod 187$$

Sub-computation	Working	Result (mod 187)
83^2	$6889 = 187 \times 36 + 157$	157
83^4	$157^2 = 24649 = 187 \times 131 + 152$	152

$7 = 4 + 2 + 1$, so:

$$M' = (83^4 \times 83^2 \times 83^1) \bmod 187 = (152 \times 157 \times 83) \bmod 187 = 8$$

Since $M' = 8 = M$, the signature is successfully verified.

Final Answer

$n = 187$, $\phi(n) = 160$, $d = 23$, Signature $S = 83$, Verification: $M' = 8 = M$ (Valid)

Analysis

Because the signature $S = 83$ could only have been produced using the signer's private key d , and verification succeeds only with the matching public key e , the signer cannot later deny having signed the message. This property is called non-repudiation. It also guarantees authenticity (the message truly originates from the claimed sender) and integrity (any alteration of M after signing would cause verification to fail). Digital signatures built on this principle underpin secure software distribution, digital contracts, and financial transactions.

Question 4 — Diffie–Hellman with Attack Resistance Evaluation

Problem Understanding

This problem requires computing the public keys of two users and the shared secret key using the Diffie–Hellman key exchange protocol, and analysing the protocol's vulnerability to interception.

- Given: $p = 29$, $g = 2$
- User A private key $a = 5$, User B private key $b = 12$
- Required: public keys A and B, and the shared secret key

Method / Formulae Used

$$A = g^a \bmod p \quad (\text{User A's public key}) \quad (1)$$

$$B = g^b \bmod p \quad (\text{User B's public key}) \quad (2)$$

$$K = B^a \bmod p = A^b \bmod p \quad (\text{Shared secret key}) \quad (3)$$

Solution

Step 1 — Compute User A's public key

$$A = g^a \bmod p = 2^5 \bmod 29 = 32 \bmod 29 = 3$$

Step 2 — Compute User B's public key

$$B = g^b \bmod p = 2^{12} \bmod 29$$

$$2^{10} = 1024 \bmod 29 = 1024 - 1015 = 9; \quad 2^{12} = 2^{10} \times 2^2 = 9 \times 4 = 36 \bmod 29 = 7$$

$$B = 7$$

Step 3 — User A computes the shared secret

$$K = B^a \bmod p = 7^5 \bmod 29$$

Sub-computation	Working	Result (mod 29)
7^2	$49 = 29 + 20$	20
7^4	$20^2 = 400 = 29 \times 13 + 23$	23
7^5	$23 \times 7 = 161 = 29 \times 5 + 16$	16

$$K (\text{via User A}) = 16$$

Step 4 — User B computes the shared secret (verification)

$$K = A^b \bmod p = 3^{12} \bmod 29$$

Sub-computation	Working	Result (mod 29)
3^2	9	9
3^4	$9^2 = 81 = 58 + 23$	23
3^8	$23^2 = 529 = 29 \times 18 + 7$	7

3^{12}	$3^8 \times 3^4 = 7 \times 23 = 161 = 145 + 16$	16
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K (via User B) = 16 — identical to Step 3, confirming both parties derive the same shared secret.

Final Answer

Public keys: A = 3, B = 7, Shared Secret Key K = 16

Analysis

If an attacker intercepts A and B in transit and substitutes fraudulent values of their own, both User A and User B will unknowingly establish separate shared secrets with the attacker instead of with each other. The attacker can then decrypt, read, and re-encrypt every message passed between them, an attack known as the man-in-the-middle attack. Plain Diffie–Hellman provides no mechanism to authenticate the origin of A and B.

- Cryptographic solution: authenticate the exchanged public keys using digital signatures or certificates issued by a trusted Certificate Authority (as in station-to-station protocol or TLS with certificate-based Diffie–Hellman), so any tampering is detectable before the shared key is used.

Question 5 — Hybrid Encryption System Analysis

Problem Understanding

This problem compares the total time taken by a hybrid encryption system (RSA for key exchange, AES for bulk data) against a system that uses RSA alone to encrypt all data blocks.

- Given: RSA encryption time = 5 ms per key exchange (performed once)
- AES encryption time = 0.5 ms per data block
- Total data blocks = 500
- Required: total hybrid encryption time, comparison with RSA-only encryption, and efficiency analysis

Method / Formulae Used

$$T(\text{hybrid}) = T(\text{RSA key exchange}) + (T(\text{AES per block}) \times N) \quad (1)$$

$$T(\text{RSA-only}) = T(\text{RSA per block}) \times N \quad (2)$$

$$\text{Speed-up factor} = T(\text{RSA-only}) / T(\text{hybrid}) \quad (3)$$

Solution

Step 1 — Compute total hybrid encryption time

$$T(\text{hybrid}) = T(\text{RSA key exchange}) + (\text{AES time per block} \times \text{number of blocks})$$

$$T(\text{hybrid}) = 5 + (0.5 \times 500) = 5 + 250 = 255 \text{ ms}$$

Step 2 — Compute total time using RSA only for all data blocks

$$T(\text{RSA-only}) = \text{RSA time per block} \times \text{number of blocks}$$

$$T(\text{RSA-only}) = 5 \times 500 = 2500 \text{ ms}$$

Step 3 — Compare the two systems

Approach	Computation	Total Time
Hybrid (RSA + AES)	$5 + (0.5 \times 500)$	255 ms
RSA only (all blocks)	5×500	2500 ms

Step 4 — Compute the speed-up factor

$$\text{Speed-up} = T(\text{RSA-only}) / T(\text{hybrid}) = 2500 / 255 \approx 9.8$$

The hybrid approach is approximately 9.8 times faster than encrypting all data blocks with RSA alone.

Final Answer

$T(\text{hybrid}) = 255 \text{ ms}$, $T(\text{RSA-only}) = 2500 \text{ ms}$, Hybrid system is $\approx 9.8\times$ faster

Analysis

RSA is computationally expensive and works on small fixed-size blocks, whereas AES is a lightweight symmetric algorithm well suited to encrypting large volumes of data quickly. Hybrid encryption uses RSA only once, to securely exchange a session key, and then uses fast AES for the actual data, combining RSA's strong key-distribution security with AES's high throughput. This is why hybrid encryption is used in nearly all modern secure communication protocols, including TLS/SSL, PGP, and secure messaging systems, where both strong security and practical performance are essential.

